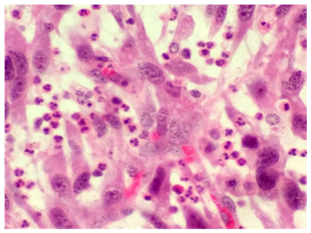
Acute Inflammation: Answer Sheet

Practical + Flipped Case-Based Learning (Darla) · answers based on the lecture notes
MDP10906 Paraclinical Sciences · 2026/2027

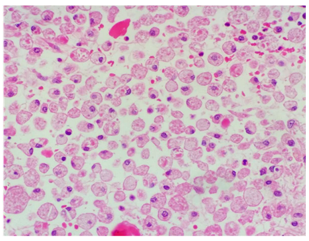
Note: the slides give the questions only, so these answers are written from the Robbins lecture note. They are in exam style (key words in **bold**). For the numbered labels 1–6 on slides I and J, I read them from the pictures, so please confirm them in class.

PART A · PRACTICAL

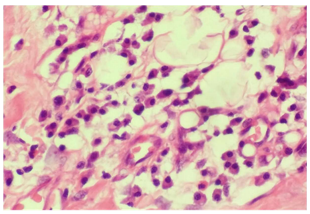
A. Identify inflammatory cells (pictures A–D)



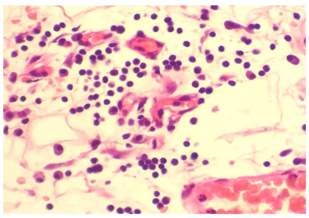
A



B



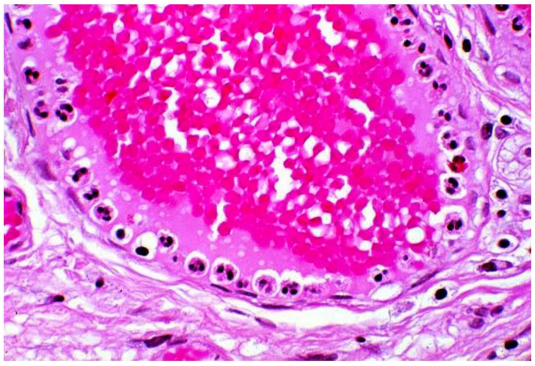
C



D

Pic	Cell	Identifying features	Predominant in (clinical conditions)
A	**Neutrophils**	Small cells; nucleus with **2–5 lobes** (multilobed); pale pink cytoplasm; scattered among larger tissue cells	**Acute inflammation**: bacterial infections (e.g. appendicitis, bronchopneumonia, abscess, cellulitis), tissue necrosis (e.g. early myocardial infarction). First cell to arrive (6–24 h).
B	**Macrophages**	**Large** cells; abundant pale/foamy cytoplasm; round to oval (kidney-shaped) nucleus	Replace neutrophils at 24–48 h; **chronic inflammation**, granulomatous disease (e.g. TB); clearing debris and microbes; healing/repair
C	**Plasma cells**	Round/oval cell; **eccentric nucleus** with clock-face chromatin; basophilic cytoplasm with perinuclear clear zone (Golgi)	**Chronic inflammation** and chronic infections (e.g. syphilis); produce **antibodies**
D	**Lymphocytes**	**Small**, round, dense dark round nucleus; very thin rim of cytoplasm	**Chronic inflammation**; **viral infections**; autoimmune disease; chronic infections (TB); normal in appendix submucosa

B. Stage of cellular event (picture E)



E: venule filled with red cells; white cells along the wall

3 stages present in E:

- **Margination (and rolling):** neutrophils move from the centre of the vessel to the periphery and line up along the endothelium after slowing of flow (stasis).
- **Adhesion:** neutrophils bind firmly to the activated endothelium (integrins binding ICAM-1/VCAM-1).
- **Transmigration (diapedesis):** neutrophils squeeze through the endothelium and basement membrane into the surrounding tissue (then chemotaxis toward the injury).

Case study 1: Acute inflammation in bronchopneumonia

(i) Gross morphology: F (pneumonia) vs G (normal)



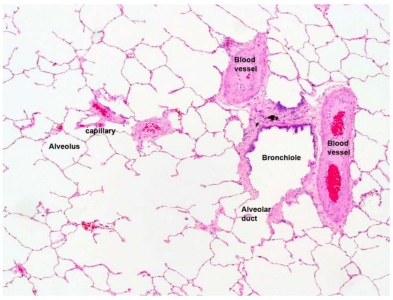
F: pneumonia (arrows = consolidation)



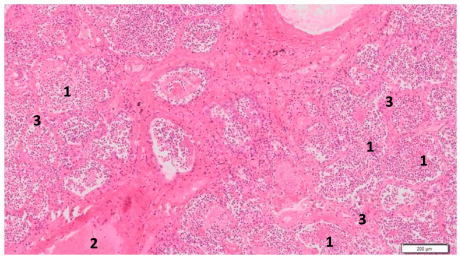
G: normal lung

Feature	G: normal lung	F: bronchopneumonia
Appearance	Uniform brown-pink, **spongy**, air-filled	**Patchy consolidated areas** (arrows): pale yellow-tan to grey, **firm and solid**
Distribution	Even	Patchy, centred around bronchi/bronchioles (bronchopneumonia pattern)
Meaning	Air in alveoli	Alveoli filled with **inflammatory exudate** (neutrophils, fluid), replacing air

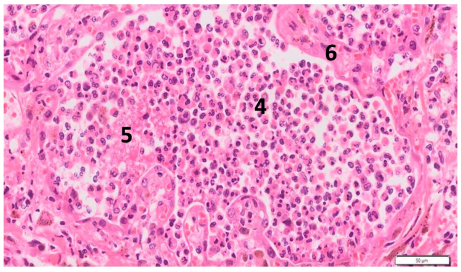
(ii) Microscopy: H (normal), I and J (bronchopneumonia)



H: normal lung



I: low power



J: high power

Task 1: features of acute inflammatory exudate in bronchopneumonia

- **Alveoli and bronchioles are filled** with exudate instead of air.
- The exudate contains **many neutrophils (pus cells)**, with some macrophages.
- **Pink protein-rich oedema fluid** (and fibrin) between the cells.
- **Alveolar septa are thickened** and the **capillaries are congested/dilated** (vascular changes).

Task 2: blood vessels and neutrophils

- Blood vessels and septal capillaries are **dilated and congested** with red cells.
- Neutrophils are **lined up along the vessel walls (margination)**, are seen **crossing the wall (emigration)** and are **concentrated around the vessels** and in the adjacent alveoli.

Task 3: normal lung (H)

- Alveoli are empty/air-filled with **thin septa**; vessels are not congested.
- Neutrophils are **few and only within the capillaries/blood vessels** (circulating); none in the alveolar spaces.

Numbered structures in I and J

No.	Identification	Slide
1	**Alveoli filled with inflammatory exudate** (neutrophils and fluid)	I
2	**Blood vessel** (large vessel with red cells/pink content)	I
3	**Thickened, congested alveolar septa** (walls between exudate-filled alveoli)	I
4	**Neutrophils** (polymorphs) packing the alveolar lumen	J
5	**Pink proteinaceous oedema fluid / fibrin** with cell debris in the alveolar lumen	J
6	**Alveolar wall with congested capillaries**	J

Case study 2: Acute appendicitis

Case: 12-year-old boy; fever, nausea, abdominal pain 2 days; pain began **around the umbilicus** and **shifted to the right lower abdomen**; tender over the right iliac fossa. Diagnosis: **acute (suppurative) appendicitis**.

(i) Gross pathology: K (patient) vs L (normal)



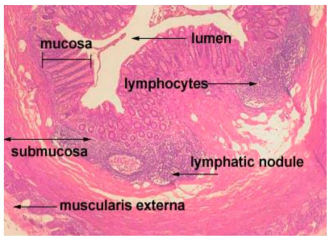
K: patient's appendix



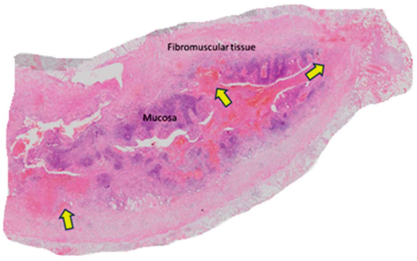
L: normal appendix

Feature	L: normal	K: acute appendicitis (3 gross features)
i. Vascular changes	Thin, pale pink, fine vessels	**Congested, dilated serosal blood vessels** (red, haemorrhagic)
ii. Swelling	Slim, small calibre	**Swollen, enlarged, thickened** appendix (oedema)
iii. Exudate	Smooth, glistening serosa	**Dull, yellow-green purulent/fibrinous exudate** (pus) on the serosal surface

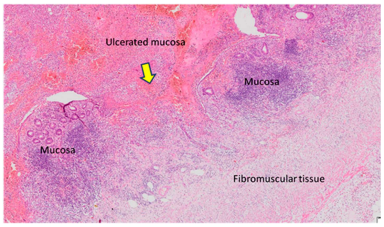
(ii) Microscopy: M (normal) and N, O, P (patient)



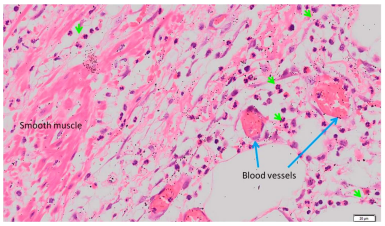
M: normal appendix



N



O



P

M, normal appendix: intact mucosal lining with crypts and a lumen; **submucosal lymphocytes** and lymphoid nodules; **muscularis externa** (smooth muscle) outside.

Observe (N, O, P)	Findings
i. Mucosal lining	**Ulcerated mucosa**: lining is lost/eroded (arrow in O); crypt architecture destroyed
ii. Inflammatory cells	Dense infiltrate of **neutrophils** through mucosa, submucosa and the muscle wall (fibromuscular tissue); neutrophils around vessels (green arrows in P)
iii. Necrosis	**Necrotic (pink, structureless) areas** of mucosa and wall; with oedema and haemorrhage

4 microscopic features of acute inflammation (N, O, P):

- **i. Mucosal ulceration/necrosis:** loss of the mucosal lining with necrotic debris.
- **ii. Neutrophil infiltration** (acute inflammatory cells) throughout the wall; pus formation (suppurative).
- **iii. Vascular changes:** dilated, congested blood vessels (blue arrows in P), with neutrophils marginating and emigrating.
- **iv. Oedema (exudate)** separating the muscle fibres (smooth muscle in P), with haemorrhage; transmural spread.

PART B · FLIPPED CASE-BASED LEARNING: DARLA

Case: Darla, 25, **cat bite** on the right middle finger. Within 24 h: skin around the puncture wound reddened and swollen; **PIP joint extremely painful**; partial loss of finger flexion. Temperature **38 °C**; swollen finger; skin at the wound **warmer** than the opposite hand. Bite site likely contaminated by bacteria. She was given symptom-relief medication and antibiotics.

Q1. State the pathological process

Acute inflammation, caused by bacterial infection of the cat-bite puncture wound (an acute suppurative/pyogenic inflammation of the finger).

Q2. Define it

Acute inflammation is a **rapid, short-lived protective response** of vascularised tissue to infection or injury. It delivers **leukocytes and plasma proteins** (antibodies, complement) from the blood to the site, to **eliminate the cause**, remove damaged tissue and **start repair**. It develops in minutes to hours, lasts hours to a few days and is characterised by **fluid and plasma protein leakage (oedema)** and **neutrophil emigration**.

Q3. Cardinal signs seen in Darla's hand

Cardinal sign	Evidence in Darla
Redness (rubor)	Skin around the puncture wound reddened
Swelling (tumor)	Swollen finger and surrounding tissue
Heat (calor)	Skin at the wound warmer than the same site on the opposite hand
Pain (dolor)	PIP joint extremely painful
Loss of function (functio laesa)	Partially lost the ability to flex her finger

Systemic feature: fever (38 °C) from IL-1, TNF and IL-6.

Q4. Pathophysiology of the cardinal signs

Sign	Pathophysiology
Redness and heat	**Histamine** (and other mediators) relaxes **arterioles** (precapillary sphincters) → vasodilation, opening of capillary beds → **↑ blood flow** to the area → redness and heat.
Swelling	Histamine (and bradykinin, leukotrienes) makes **venules more permeable** (endothelial contraction) → **protein-rich fluid leaks** into the interstitial space → **oedema**.
Pain	**Prostaglandin E2 (PGE2)** makes nerve endings more sensitive to **bradykinin** and other pain mediators; swelling also stretches/presses on nerves.
Loss of function	Result of pain, swelling and tissue injury.

Q5. Pathophysiologic categories of oedema

Oedema = excess fluid in tissue (effusion = in body cavities). Normally hydrostatic pressure pushes fluid out of capillaries and colloid osmotic pressure pulls it back in; lymphatics drain the excess. Oedema forms when this balance is disturbed. It is either **inflammatory (exudate)** or **non-inflammatory (transudate)**.

Category	Mechanism	Examples
1. ↑ Hydrostatic pressure	Impaired venous return → more fluid pushed out	Congestive heart failure, DVT, venous obstruction, constrictive pericarditis
2. ↓ Plasma osmotic pressure	Low albumin (↓ synthesis or ↑ loss) → less fluid drawn back	Liver cirrhosis, nephrotic syndrome, malnutrition, protein-losing enteropathy
3. Na^+ and water retention	↑ blood volume (↑ hydrostatic) + dilution (↓ osmotic)	Renal dysfunction, excess salt intake, ↑ RAAS
4. Lymphatic obstruction	Interstitial fluid cannot drain → lymphoedema	Trauma, fibrosis, tumour, infection, surgery, radiation
5. Inflammation	**↑ vascular permeability** → protein-rich exudate escapes	Acute/chronic inflammation, burns, infection

Darla = category 5 (inflammatory oedema, an exudate). Exudate: protein-rich, contains cells and debris. Transudate: protein-poor, no inflammation.

Q6. Causes that can stimulate acute inflammation

Cause	Examples	In Darla
1. Infections (most common)	Bacteria, viruses, fungi, parasites, toxins	**Yes**: bacteria from cat's mouth/claws
2. Tissue necrosis	Ischaemia, trauma, burns, frostbite, radiation, chemicals	Puncture trauma/dead cells at the wound
3. Foreign bodies	Splinters, dirt, sutures; urate and cholesterol crystals	Possible (debris/teeth fragments)
4. Immune reactions	Autoimmune (rheumatoid arthritis), allergy (pollen, food), excessive responses	No

Q7. Vascular and cellular events

Vascular events

- **Brief vasoconstriction** of arterioles (neurogenic reflex, seconds).
- **Vasodilation** of arterioles and opening of new capillary beds (histamine, prostaglandins, NO, kinins) → ↑ blood flow → redness and heat.
- **↑ Vascular permeability of postcapillary venules** (hallmark): endothelial contraction (histamine, bradykinin, leukotrienes; immediate transient, 15–30 min) and endothelial injury (microbial toxins, neutrophil injury; prolonged).
- **Exudation** of protein-rich fluid → oedema (swelling).
- Fluid loss + dilation → slow flow, ↑ viscosity → **stasis** → neutrophil **margination**.
- Lymphatics: ↑ lymph flow to drain oedema, microbes and debris; possible **lymphangitis/lymphadenitis**.

Cellular events (neutrophils first, 6–24 h)

Step	What happens	Molecules
1. Margination	WBCs move to the vessel wall	Stasis
2. Rolling	Loose, transient binding to endothelium	Selectins (E-, P-selectin; L-selectin)
3. Adhesion	Firm binding to endothelium	Integrins ↔ ICAM-1, VCAM-1 (induced by TNF, IL-1)
4. Transmigration (diapedesis)	Cross the endothelium (postcapillary venule) and basement membrane	CD31 (PECAM-1), collagenases
5. Chemotaxis	Migrate along a chemical gradient to the bite site	Bacterial products, IL-8, C5a, LTB4
6. Activation and phagocytosis	Recognise and kill bacteria; macrophages replace neutrophils at 24–48 h	TLRs, opsonins (IgG, C3b)

Q8. Chemical mediators and their actions

Mediator	Source	Action
Histamine	Mast cells (also basophils, platelets)	Arteriolar vasodilation; ↑ venular permeability (H1) → redness, heat, swelling; immediate phase
Serotonin	Platelets	Vasoconstriction; minor role
Prostaglandins (PGE2, PGD2, PGF2α, PGI2)	COX pathway (mast cells, leukocytes, endothelium)	PGE2: **pain, fever**, vasodilation. Others: vasodilation, ↑ permeability. TxA2: platelet aggregation, vasoconstriction
Leukotrienes	5-LOX pathway (neutrophils, macrophages, mast cells)	LTB4: **neutrophil chemotaxis**. LTC4, D4, E4: ↑ permeability, vasoconstriction, bronchoconstriction
Lipoxins	Neutrophils, platelets	Anti-inflammatory; stop neutrophil recruitment; resolution
TNF, IL-1	Macrophages, dendritic cells	Endothelial activation (↑ selectins, ICAM-1, VCAM-1); leukocyte activation; **fever**; acute-phase response
IL-6	Macrophages	Fever, acute-phase proteins, leukocytosis
Chemokines (IL-8)	Macrophages, endothelium	**Neutrophil chemotaxis** and activation; ↑ integrin affinity
Complement (C3a, C5a, C3b, MAC)	Plasma (liver)	C3a/C5a: ↑ permeability, mast cell degranulation; **C5a: chemotaxis**; **C3b: opsonin**; MAC: lysis of bacteria
Bradykinin	Plasma kininogen (via kallikrein)	↑ Permeability, vasodilation, **pain**
PAF	Platelets, mast cells, leukocytes, endothelium	Vasodilation, very potent ↑ permeability, leukocyte activation

Mediator	Source	Action
Neuropeptides (substance P)	Sensory nerves	Pain, ↑ permeability

Q9. Phagocytosis of the bacteria

Done mainly by **neutrophils and macrophages**:

- **1. Recognition and attachment:** receptors (mannose, scavenger, TLRs) bind the bacterium; coating by **opsonins (IgG, C3b, mannose-binding lectin)** enhances binding.
- **2. Engulfment:** pseudopods surround the bacterium → **phagosome** → fuses with a **lysosome** → **phagolysosome**.
- **3. Killing and degradation: oxygen-dependent** (reactive oxygen species, myeloperoxidase–H_2O_2–halide = hypochlorite) and **oxygen-independent** (lysosomal enzymes, nitric oxide) mechanisms, all inside the lysosome to protect the cell. Neutrophils may also release **NETs**. Leakage of enzymes/ROS may damage nearby tissue.

Q10. Most likely morphological pattern

Purulent (suppurative) inflammation: pus = neutrophils + necrotic debris + oedema fluid, produced by pyogenic bacteria (e.g. staphylococci) causing liquefactive necrosis. If it becomes localised it forms an **abscess** (central liquefied necrosis with neutrophils; surrounding viable neutrophils; outer rim of dilated vessels and fibroblasts).

Q11. Possible outcomes

Outcome	Explanation (applied to Darla)
1. Complete resolution	Most likely with prompt antibiotics: cause removed; macrophages clear dead cells; oedema drains via lymphatics; tissue regenerates; finger returns to normal
2. Healing by fibrosis (scar)	If tissue destruction or an abscess forms (tissue cannot fully regenerate): fibroblasts grow in (organisation) → scar → stiff finger, reduced flexion
3. Progression to chronic inflammation	If bacteria persist or healing is disturbed: ongoing injury and repair (e.g. chronic infection of the joint)

Complications if untreated: spread via lymphatics (lymphangitis, lymphadenitis) or to the joint (septic arthritis of the PIP joint) and systemic spread (sepsis).

Q12. Prescribed medications, with reasons

Medication	Reason
Antibiotics	Treat the **cause**: kill the bacteria introduced by the bite, so inflammation can resolve and spread to the joint/blood is prevented. (Cat-bite bacteria need broad coverage, e.g. amoxicillin-clavulanate; use the example from your lecturer.)
NSAIDs (e.g. ibuprofen, naproxen; aspirin)	**Relieve symptoms**: block **COX-1 and COX-2** → ↓ **prostaglandins** (PGE2) → ↓ **pain, fever and swelling/inflammation**.
Paracetamol (alternative)	Reduces **pain and fever** (little anti-inflammatory effect).

Corticosteroids (↓ phospholipase A2, cytokines) also suppress inflammation but are not needed in a simple infection, since the aim is to let the body fight the bacteria.